

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457371.77142 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.046
Transit depth (Rp/Rs)^2	2.4 +/- 1.42 [%]
Orbital Inclination (inc)	86.4 +/- 2.93
Airmass coefficient 1 (a1)	1.38 +/- 0.73
Airmass coefficient 2 (a2)	-0.34 +/- 0.85
Scatter in the residuals of the lightcurve fit is	55.42 %
Best Comparison Star	#2 - [185, 177]
Optimal Aperture	2.76
Optimal Annulus	4.75
Transit Duration (day)	0.1226 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.046 vs 0.1178 (deviation 0.81σ)
Duration fit vs published	0.1226 d vs 0.125 d (deviation 0.69σ)
Mid-time O–C	-1571.8 min
Depth SNR	3.4
Residual scatter	55.42 %

Light curve

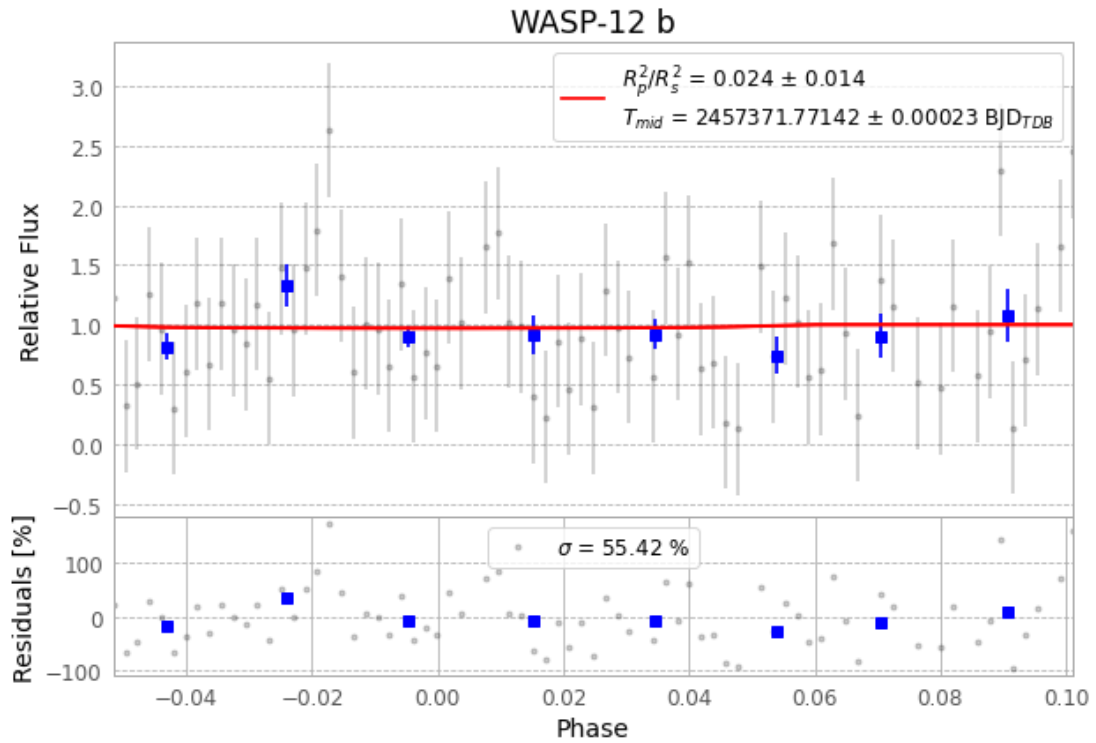


Figure 1 — FinalLightCurve_WASP-12 b_2015-12-14.png (SHA-256 dc433ebf57a14b7a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [470, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5f1999709fbd3071...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

84 FITS frames (55 MB), WASP-12151215050012.FITS → WASP-12151215090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-151216.log (997c9be8b2f3...), FinalLightCurve_WASP-12 b_2015-12-14.png (dc433ebf57a1...), FinalLightCurve_WASP-12 b_2015-12-14.csv (0028b1edc71c...)

Compute resources

Wall time 123.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:32Z.